Priyanshu Shekhar

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Education

University of Southern California (USC) — **June 2024 – May 2026**
Master of Science in Computer Science (Specialization: Artificial Intelligence) — *Los Angeles, California*
Relevant Coursework: Analysis of Algorithms, Foundations of AI, Machine Learning Systems, Deep Learning and Its Applications, Applied Natural Language Processing, Robotics, Multi-modal Probabilistic Learning of Human Communication, Web Technologies

California State Polytechnic University-Pomona (Cal Poly Pomona) — **August 2021 – May 2024**
Bachelor of Science in Computer Science (Minor: Data Science) | Summa Cum Laude — *Pomona, California*
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Systems, Distributed Systems, Probability and Statistics Big Data Analytics & Cloud Computing, Web Search & Recommender Systems, Systems Programming

Professional Experience

Research Assistant — **October 2025 – May 2026**
SentryFuzz Project, Lime Lab, USC — *Los Angeles, California*
- Designed Python-based evaluation frameworks for Large Language Models (LLMs), implementing multi-step testing workflows that validated AI-generated outputs across complex privacy attack scenarios
- Designed graph-based synthetic user profiles spanning 3+ sensitive domains (financial, medical, and personal), enabling retrieval-driven evaluation of information extraction attacks and responsible AI testing
- Implemented evaluation strategies from 25+ AI security and adversarial ML publications, improving testing coverage and guiding architecture decisions for scalable AI evaluation services
- Collaborated with researchers to prototype, test, and iterate AI evaluation, supporting deployment-ready backend services through experimentation

AI Engineer Intern — **February 2024 – May 2024**
Google ExploreCSR Project, Prof. Eger's Lab, Cal Poly Pomona — *Pomona, California*
- Architected Python backend services, REST APIs, and MySQL data pipelines supporting real-time analysis of 50,000+ records through scalable application services
- Developed a full–stack application using React, and backend APIs, designing modular software components that simplified feature integration and improved maintainability
- Built SQL and Pandas analysis pipelines that improved reporting accuracy while supporting responsible AI evaluation and data-driven product decisions
- Collaborated with cross-functional teams throughout the software development lifecycle, balancing implementation trade-offs across design, testing, deployment, and continuous improvement

Machine Learning Engineer — **May 2023 – February 2024**
Agricultural Precision Project, Prof. Korah's Lab, Cal Poly Pomona — *Pomona, California*
- Reimplemented the STARFM geospatial forecasting algorithm as USTARFM in Python, improving prediction accuracy by 15% through optimized algorithm implementation for satellite imagery
- Designed distributed Python pipeline on AWS EC2 to process 300+ satellite image dataset, supporting scalable model training & cloud-based inference
- Built feature engineering and preprocessing pipelines for large-scale geospatial datasets, improving model performance by 25%
- Partitioned workloads between edge devices and cloud infrastructure, reducing processing latency by 35% and improving system scalability and operational efficiency

Personal Projects

ASLTutor – Multimodal ASL Learning Platform — **January 2026 – May 2026**
- Built an AI–native learning platform using FastAPI, React, Whisper ASR, and LLM that delivered real-time multimodal ASL tutoring via backend APIs
- Designed a computer vision workflow combining MediaPipe and Graph Neural Networks to generate adaptive feedback through AI inference
- Implemented prompt engineering, Whisper ASR, and Transformer models to convert speech into ASL gloss sequences with low-latency inference
- Optimized backend APIs and inference orchestration to improve response latency while maintaining production-ready software architecture

Event Finder Web and Android Application — **September 2025 – January 2026**
- Built and shipped a full-stack event discovery application using React, TypeScript, Node.js, and Google Cloud that aggregated live event data from multiple external APIs
- Designed and deployed microservices using Node.js, REST APIs, and Google Cloud to integrate multiple event providers, enabling modular feature development and scalable backend services
- Designed MongoDB data models and aggregation pipelines that supported reliable data exchange between microservices, improving backend scalability and system maintainability
- Optimized backend query execution and API workflows to improve application responsiveness and reduce operational overhead

SpamNet Web Application — **May 2025 – August 2025**
- Developed a Python Flask backend that automated real-time spam detection using PyTorch, achieving 89% classification accuracy
- Built REST APIs integrating backend inference services with a TypeScript frontend, enabling AI-powered moderation workflows
- Designed model evaluation and system testing that improved deployment reliability while validating prediction accuracy before production use
- Integrated the YouTube Data API to automate continuous ingestion and analysis of user-generated content for near real-time spam detection

Technical Skills

Programming Languages: Python, Java, JavaScript, C, SQL, Bash
Web Development: React, Node.js, Flask, Django, FastAPI, RESTful APIs, TailwindCSS
Data Management: MySQL, MongoDB, SQLite, Hadoop
AI & ML Frameworks: GitHub Copilot, Prompt Engineering, PyTorch, TensorFlow, Scikit-learn, Hugging Face, RAG, AI Agents, NumPy, Pandas
Cloud & Infrastructure: AWS (EC2), GCP, Docker, Kubernetes, Git, Microservices Architercture